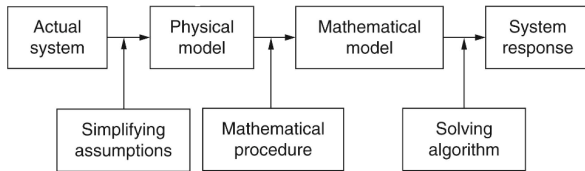


Controls and Dynamics

Modelling

Procedure



Elements and Equations

$$e_m = k_e \omega \quad T_m = k_t l$$

$$T_{\text{stall}} = T_a|_{\omega=0, \dot{\omega}=0} \quad \omega_0 = \omega|_{T_a=0, \dot{\omega}=0}$$

Extended FBD Method

1. Divide physical model into ideal elements
2. Assign force and displacement references
3. Model single elements using assigned references
4. Model the interactions between the elements using D'Alembert Law

$$\Sigma \mathbf{F}_{\text{ext}} - M \frac{d\mathbf{v}}{dt} = 0$$

$$\mathbf{F}_m = M\ddot{x} \quad \uparrow \downarrow \text{ to positive disp.}$$

5. Substitute element equations into D'Alembert Law equation(s)
6. Algebraic equations into standard form

Electrical Systems

1. Assign current and voltage references to each element
2. Identify loops and nodes
3. Model single elements using assigned references
4. Use KCL, and KVL on loops and nodes
5. Combine equations towards ODE(s)

State Space

1. Define state space variables vector $\mathbf{q}$. Pick variables that deal with energy, such as:

- x/θ - potential energy
- v/ω - kinetic energy
- e_c - energy in a capacitor
- l_L - energy in an inductor

2. Apply the derivative, and form the following equations:

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{q} + \mathbf{D}\mathbf{u}$$

Where:

- A, B, C, D are matrices
- $\mathbf{u}$ is the input forces vector
- $\mathbf{y}$ is some force in the system (i.e. force on a wall)

3. Solve using numeric solver (MATLAB/Simulink)

ODE General Forms

$$\dot{x}(t) + \frac{1}{\tau}x(t) = \kappa u(t)$$

Where:

- τ is the time constant
- κ is the (input) sensitivity

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2x(t) = \kappa\omega_n^2u(t)$$

Where:

- ζ is the damping ratio
 - ω_n is the natural frequency
-

Laplace Land

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \\ = \alpha \pm \beta$$

$$\tau = -\frac{1}{\alpha} \quad T_s = 4\tau \quad T_{\text{peak}} = \frac{\pi}{\beta} \quad O.S. = \exp \frac{\pi\alpha}{\beta}$$

- Overdamped ($\zeta > 1$)

$$PFE \rightarrow \frac{A}{s} + \frac{B}{s - s_1} + \frac{C}{s - s_2}$$

- Critically Damped ($\zeta = 1$)

$$PFE \rightarrow \frac{A}{s} + \frac{B}{(s - s_1)^2} + \frac{C}{s - s_1}$$

- Underdamped ($0 < \zeta < 1$)

$$PFE \rightarrow \frac{A}{s} + \frac{Bs + C}{(s - \alpha)^2 + \beta^2}$$

- No damping ($\zeta = 0$)

Linearization

Taylor series approximation:

$$f(x)|_{x=\bar{x}} = f(x)|_{x=\bar{x}} + \left. \frac{\partial f(x)}{\partial x} \right|_{x=\bar{x}} (x - \bar{x})$$

Controls

$$y_{ss} = \lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow \infty} sY(s)$$

$$G(s) = \frac{Y(s)}{U(s)} \quad G(s)_{\text{closed}} = \frac{G(s)_{\text{open}}}{1 + G(s)_{\text{open}}}$$

Controllers

P	$C(s) = k_p$
PD	$C(s) = k(s - z_0)$
PI	$C(s) = k \frac{(s - z_0)}{s}$
PID	$C(s) = k \frac{(s - z_0)(s - z_1)}{s}$

Root Locus

1. RL begins at the poles of the O.L.T.F. and ends at the holes of the O.L.T.F.
2. The # of poles = the # of branches. Every pole moves towards a zero (finite or infinite)
3. RL is symmetric about the real-axis
4. Left of odd poles/holes. Moving right to left, count the real-axis poles and holes the RL exists to the left of odd numbered entries
5. Asymptotes

$$\#A = \#_{\text{finite poles}} - \#_{\text{finite zeros}}$$

$$\sigma_A = \frac{\sum s_i - \sum z_j}{\#A} \quad \theta_{A_i} = \frac{(2m + 1)\pi}{\#A}$$

$$m = 0, 1, 2, \dots (\#A - 1)$$

6. Break away/in points

- (a) Find the O.L.T.F., then the char. eq.

- (b) Solve for k

- (c) Set $s = \alpha \pm j0$, find $k(\alpha) = eq$.

- (d) Find alpha value(s) that satisfy $\frac{\partial k}{\partial \alpha} = 0$

- (e) Check them!

7. Stability-Instability points

- (a) Find the O.L.T.F., then the char. eq.

- (b) Input $s = 0 \pm j\beta$

- (c) Solve the Re, Im parts, achieve k value(s)

- (d) Check them!